

Homework 02

Question 1: Training and test error under a fixed design

Consider a linear regression problem with $n = 100$ observations and $p = 20$ available covariates. Generate one matrix $\mathbf{X}_{\text{all}} \in \mathbb{R}^{n \times p}$ whose entries are independent $\mathcal{N}(0, 1)$ random variables, and then keep this realized matrix fixed. Let

$$\beta_j = 0.4^{\sqrt{j}}, \quad j = 1, \dots, p,$$

and define

$$\boldsymbol{\mu} = \mathbf{X}_{\text{all}}\boldsymbol{\beta}.$$

Generate one training response and one independent test response from

$$\begin{aligned} \mathbf{y} &= \boldsymbol{\mu} + \boldsymbol{\epsilon}, \\ \mathbf{y}^* &= \boldsymbol{\mu} + \boldsymbol{\epsilon}^*, \end{aligned}$$

where

$$\boldsymbol{\epsilon}, \boldsymbol{\epsilon}^* \stackrel{\text{ind}}{\sim} \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n).$$

Here $\mathbf{I}_n$ is the $n \times n$ identity matrix.

For $m = 0, 1, \dots, p$, fit a linear model with an intercept and the first m columns of $\mathbf{X}_{\text{all}}$ using $\mathbf{y}$. If $\hat{\mathbf{y}}_m$ is its fitted mean vector, define

$$\text{MSE}_{\text{train},m} = \frac{1}{n} \|\mathbf{y} - \hat{\mathbf{y}}_m\|_2^2,$$

and

$$\text{MSE}_{\text{test},m} = \frac{1}{n} \|\mathbf{y}^* - \hat{\mathbf{y}}_m\|_2^2.$$

Repeat this process independently 200 times, keeping $\mathbf{X}_{\text{all}}$ fixed.

Use seed [43202](#) before generating $\mathbf{X}_{\text{all}}$ and the response errors. R and Python use different random-number generators, so their exact numerical values need not agree.

- For each m , calculate the average training and test MSE over the 200 simulation runs. Plot both curves against m . Verify numerically that training MSE does not increase as predictors are added in each simulation run.

► Show the starter code

- For the model containing the first m predictors, write

$$\mathbf{X}_m = [\mathbf{1}, \mathbf{x}_1, \dots, \mathbf{x}_m], \quad \mathbf{H}_m = \mathbf{X}_m(\mathbf{X}_m^\top \mathbf{X}_m)^{-1} \mathbf{X}_m^\top.$$

Here $\mathbf{X}_m \in \mathbb{R}^{n \times (m+1)}$ and $\mathbf{H}_m \in \mathbb{R}^{n \times n}$. The model has $m + 1$ fitted coefficients, including the intercept. Define its **total squared approximation bias** as

$$B_m^2 = \|(\mathbf{I}_n - \mathbf{H}_m)\boldsymbol{\mu}\|_2^2.$$

Thus, B_m^2/n is the mean squared approximation bias.

Calculate the two theoretical expectations

$$E(\text{MSE}_{\text{train},m} \mid \mathbf{X}_{\text{all}}) = \frac{B_m^2}{n} + 1 - \frac{m+1}{n},$$

and

$$E(\text{MSE}_{\text{test},m} \mid \mathbf{X}_{\text{all}}) = \frac{B_m^2}{n} + 1 + \frac{m+1}{n}.$$

Add these expectations to your plot and compare them with the simulation averages.

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- c. Add the test MSE curve from one simulation run to the plot. Explain why this single curve is less smooth than the average test MSE. Use the approximation-bias and estimation-variance terms in the expected test MSE to explain why training MSE cannot be used by itself to select the predictor count.

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	m	train	theory_train	test	theory_test	first_test
1	0	1.3147	1.2974	1.337	1.317	1.0024
6	5	0.9806	0.9750	1.098	1.095	0.8559
11	10	0.9061	0.8992	1.119	1.119	0.8328
16	15	0.8524	0.8419	1.159	1.162	0.9206
21	20	0.7948	0.7900	1.217	1.210	0.9590

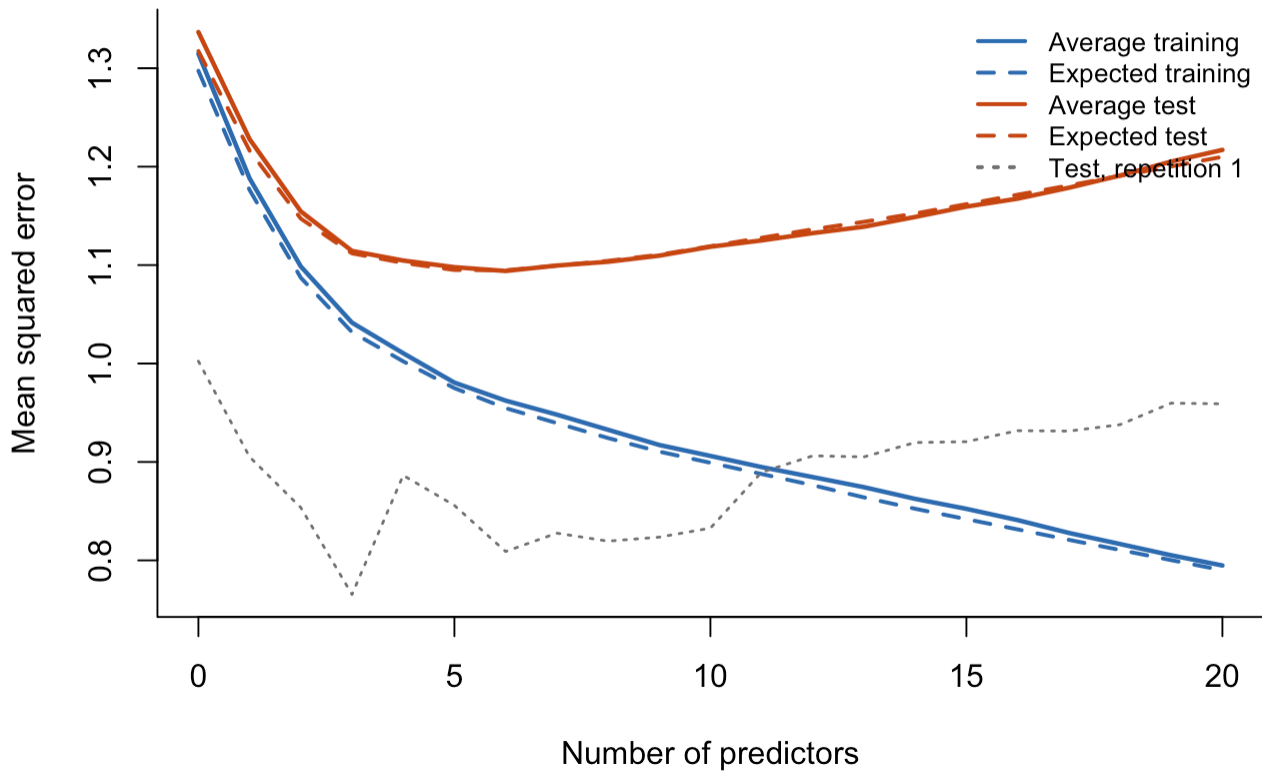
► Show the starter code

```
training_is_monotone
      TRUE
```

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```
minimum_average_test minimum_expected_test
              6              6
```

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The test-MSE curve from one run is less smooth because it is affected by the random training and test errors from that specific run. With having over 200 runs, this reduces that randomness and a smoother curve is then produced. Training MSE also cannot select the number of predictors because it continues to decrease as predictors are added. Test MSE balances decreasing the approximation bias with increasing estimation variance. Therefore, adding too many predictors worsens the test performance even while training MSE falls.

Question 2: Prediction error at one target point

Let $n = 100$, $p = 6$, and $\sigma^2 = 1$. Construct the fixed covariate matrix $\mathbf{X}_{\text{all}} \in \mathbb{R}^{n \times p}$ with entries

$$(\mathbf{X}_{\text{all}})_{ij} = \sqrt{2} \cos \left\{ \frac{\pi j (i - \frac{1}{2})}{n} \right\}, \quad i = 1, \dots, n, \quad j = 1, \dots, p.$$

Its columns satisfy

$$\mathbf{1}^\top \mathbf{X}_{\text{all}} = \mathbf{0}^\top, \quad \frac{1}{n} \mathbf{X}_{\text{all}}^\top \mathbf{X}_{\text{all}} = \mathbf{I}_p.$$

Here $\mathbf{I}_p$ is the $p \times p$ identity matrix.

The following code shows a direct construction of $\mathbf{X}_{\text{all}}$.

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Keep $\mathbf{X}_{\text{all}}$ fixed and generate one training response from

$$\mathbf{y} = \mathbf{X}_{\text{all}}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\beta} = (0.5, 0.5, 0.5, 0.5, 0.5, 0.5)^\top,$$

where

$$\boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n).$$

Here $\mathbf{I}_n$ is the $n \times n$ identity matrix.

For $m = 0, \dots, p$, fit a linear model with an intercept and the first m predictors. Consider prediction of the mean response at

$$\mathbf{x}_0 = (0, 1, 0, 0, 1, 0)^\top.$$

Write

$$\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta},$$

and let $\hat{\mu}_{0,m}$ be the prediction from the model containing the first m predictors. Here x_{0j} denotes the j th coordinate of $\mathbf{x}_0$. For this design,

$$E\left[(\hat{\mu}_{0,m} - \mu_0)^2 \mid \mathbf{X}_{\text{all}}\right] = \left(\sum_{j=m+1}^p x_{0j}\beta_j\right)^2 + \frac{\sigma^2}{n} \left(1 + \sum_{j=1}^m x_{0j}^2\right).$$

- a. Calculate μ_0 , the theoretical squared bias, and the expected squared error for $m = 0, \dots, p$ using the formula above. **You do not need to derive the formula.** Briefly explain in words why it consists of a squared bias term from omitted predictors and a variance term from estimating coefficients; no proof is required. Identify the values of m where the expected squared error changes and where it remains unchanged.

Omitted predictors create bias because their contributions are missing from the prediction. Estimating the intercept and included slopes create variance. The predictors are centered and orthogonal so the variance contributions add together.

The theoretical values are:

m	Squared target-point bias	Estimation variance	Expected squared error
0	1.00	0.01	1.01

	Squared target-point <i>m</i> bias	Estimation variance	Expected squared error
1	1.00	0.01	1.01
2	0.25	0.02	0.27
3	0.25	0.02	0.27
4	0.25	0.02	0.27
5	0.00	0.03	0.03
6	0.00	0.03	0.03

Only coordinates 2 and 5 of $\mathbf{x}_0$ are nonzero. The smallest predictor count that minimizes the expected error is $m = 5$.

b. Use seed 43203. Generate the response, fit the models, and calculate

$$(\hat{\mu}_{0,m} - \mu_0)^2.$$

Repeat this independently 200 times. Plot the simulation averages and theoretical expectations together. Report the smallest m that minimizes the theoretical error.

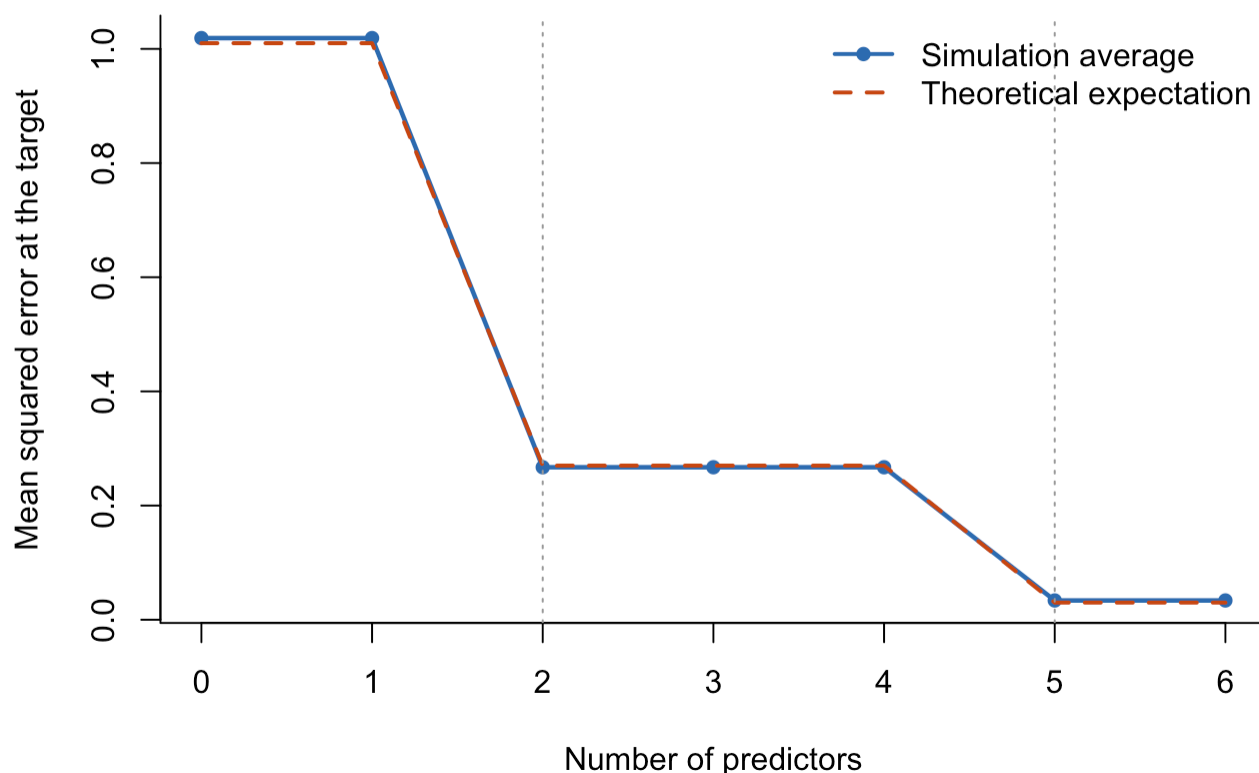
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```
m simulated theoretical
1 0 1.01892 1.01
2 1 1.01892 1.01
3 2 0.26712 0.27
4 3 0.26712 0.27
5 4 0.26712 0.27
6 5 0.03376 0.03
7 6 0.03376 0.03
```

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```
first_minimum
5
```

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The smallest predictor count that minimizes the theoretical expected error is $m = 5$.

- c. All six regression coefficients are nonzero. Explain why only predictors 2 and 5 contribute directly to the mean response at $\mathbf{x}_0$. Contrast this target-specific error with the test MSE averaged over all rows of $\mathbf{X}_{\text{all}}$ in Question 1.

Only 2 and 5 contribute directly to the mean response because they are the only coordinates that are non zero. In Question 1, test MSE was averaged across every row of the design matrix while in question 2 it evaluated error at only one specific target point. What this means is that a predictor can matter in Question 1 but really have no direct effect on a particular point.

Question 3: The optimism correction

Choose one candidate model before observing the response. Let p be its number of predictors and let $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ be its full-rank design matrix, including the intercept. The model has $p + 1$ fitted coefficients. Let

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

be its hat matrix. Suppose

$$\mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\epsilon}, \quad \mathbf{y}^* = \boldsymbol{\mu} + \boldsymbol{\epsilon}^*,$$

Conditional on the fixed design $\mathbf{X}$, the two error vectors are independent, have mean zero, and have covariance matrix $\sigma^2 \mathbf{I}_n$. Define

$$\text{MSE}_{\text{train}} = \frac{1}{n} \|\mathbf{y} - \mathbf{H}\mathbf{y}\|_2^2, \quad \text{MSE}_{\text{test}} = \frac{1}{n} \|\mathbf{y}^* - \mathbf{H}\mathbf{y}\|_2^2.$$

The conditional bias vector of the fitted mean is

$$E(\mathbf{H}\mathbf{y} \mid \mathbf{X}) - \boldsymbol{\mu} = -(\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu}.$$

Let B^2 denote the **total squared approximation bias**:

$$B^2 = \|(\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu}\|_2^2.$$

Thus, B^2/n is the mean squared approximation bias.

You may use

$$\mathbf{H}^\top = \mathbf{H}, \quad \mathbf{H}^2 = \mathbf{H}, \quad \text{tr}(\mathbf{H}) = p + 1.$$

For any fixed matrix $\mathbf{A}$, you may also use

$$E(\boldsymbol{\epsilon}^\top \mathbf{A} \boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \text{tr}(\mathbf{A}).$$

a. Derive

$$E(\text{MSE}_{\text{train}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 - \frac{p+1}{n}\right),$$

and

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 + \frac{p+1}{n}\right).$$

The training residual is

$$\begin{aligned} \mathbf{y} - \mathbf{H}\mathbf{y} &= (\mathbf{I}_n - \mathbf{H})(\boldsymbol{\mu} + \boldsymbol{\epsilon}) \\ &= (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \end{aligned}$$

The cross term has expectation zero. Since $\mathbf{I}_n - \mathbf{H}$ is symmetric and idempotent,

$$\begin{aligned} E\left(\|\mathbf{y} - \mathbf{H}\mathbf{y}\|_2^2 \mid \mathbf{X}\right) &= B^2 + \sigma^2 \text{tr}(\mathbf{I}_n - \mathbf{H}) \\ &= B^2 + \sigma^2(n - (p+1)). \end{aligned}$$

Dividing by n gives the stated expected training MSE.

For the independent test response,

$$\mathbf{y}^* - \mathbf{H}\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + \boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}.$$

All cross terms again have expectation zero. Independence gives

$$\begin{aligned} E\left(\|\epsilon^* - \mathbf{H}\epsilon\|_2^2 \mid \mathbf{X}\right) &= n\sigma^2 + \sigma^2 \text{tr}(\mathbf{H}) \\ &= \sigma^2(n + (p + 1)). \end{aligned}$$

Therefore,

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 + \frac{p + 1}{n}\right).$$

b. Deduce the expected optimism, defined as $E(\text{MSE}_{\text{test}} - \text{MSE}_{\text{train}} \mid \mathbf{X})$. For

$$n = 120, \quad p = 4, \quad \frac{B^2}{n} = 0.04, \quad \sigma^2 = 1,$$

calculate the expected training MSE, expected test MSE, and their difference. Explain why the ordering need not hold for every realized pair of responses.

Subtracting the expectations gives

$$E(\text{MSE}_{\text{test}} - \text{MSE}_{\text{train}} \mid \mathbf{X}) = \frac{2(p + 1)\sigma^2}{n}.$$

For the stated values,

$$E(\text{MSE}_{\text{train}} \mid \mathbf{X}) = 0.04 + 1 - \frac{4 + 1}{120} = 0.998333,$$

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = 0.04 + 1 + \frac{4 + 1}{120} = 1.081667,$$

and their difference is 0.083333. These are long-run averages, so a particular test MSE can still be smaller than its corresponding training MSE.

c. Continue with $n = 120$ and $p = 4$. Suppose the candidate model has residual sum of squares $\text{RSS} = 116.4$, and suppose a common estimate of the error variance is $\hat{\sigma}^2 = 0.96$.

Calculate

$$\widehat{\text{MSE}}_{\text{test}} = \frac{\text{RSS} + 2(p + 1)\hat{\sigma}^2}{n},$$

and

$$C_p = \frac{\text{RSS}}{\hat{\sigma}^2} - n + 2(p + 1).$$

Show how these two quantities are related, and explain why minimizing C_p is equivalent to minimizing the corrected test MSE when n and $\hat{\sigma}^2$ are common to all candidate models.

The corrected estimate is

$$\widehat{\text{MSE}}_{\text{test}} = \frac{116.4 + 2(4 + 1)(0.96)}{120} = 1.05.$$

Mallows' C_p is

$$C_p = \frac{116.4}{0.96} - 120 + 2(4 + 1) = 11.25.$$

Their relationship is

$$\begin{aligned} \frac{\hat{\sigma}^2(C_p + n)}{n} &= \frac{0.96(11.25 + 120)}{120} \\ &= 1.05 \\ &= \frac{\text{RSS} + 2(p + 1)\hat{\sigma}^2}{n}. \end{aligned}$$

When n and $\hat{\sigma}^2$ are common to all candidates, this is an increasing transformation. The two criteria therefore rank the models in the same order.

Question 4: Comparing Mallows' C_p , AIC, and BIC

Use `data/diabetes.csv`, with `y` as the response. Use rows 1 through 370 as the training data, and let $n = 370$ denote the training sample size. Do not use rows 371 through 442 in this question. Fit the following ordinary least-squares models, each with an intercept.

Model	Predictors	p
Model A	<code>bmi</code> , <code>bp</code> , <code>s5</code> , <code>sex</code> , <code>s1</code> , <code>s2</code> , <code>s4</code>	7
Model B	all predictors in Model A, followed by <code>s6</code>	8
Full reference	all ten predictors	10

Here p counts predictors, so each model has $p + 1$ fitted coefficients including the intercept. Estimate one common noise variance from the full reference model, using $p = 10$:

$$\hat{\sigma}^2 = \frac{\text{RSS}_{\text{full}}}{n - (p + 1)}.$$

For Models A and B, calculate

$$C_p = \frac{\text{RSS}}{\hat{\sigma}^2} - n + 2(p + 1),$$

$$\text{AIC}^{\text{red}} = n \log \left(\frac{\text{RSS}}{n} \right) + 2(p + 1),$$

and

$$\text{BIC}^{\text{red}} = n \log \left(\frac{\text{RSS}}{n} \right) + (p + 1) \log(n).$$

The reduced AIC and BIC omit constants shared by the two candidate models. Smaller values are preferred within each criterion.

- a. Fit the three models. Report p , RSS, the residual degrees of freedom of the full model, and $\hat{\sigma}^2$.

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```
[1] TRUE
```

► Show the starter code

```
[1] 370 11
```

► Show the starter code

Model A has $p = 7$ and an RSS of 1,098,908.57. Model B has $p = 8$ and an RSS of 1,089,717.06. The full model has $p = 10$, and an RSS of 1,089,451.72. The full model has 359 residual degrees of freedom because it estimates 11 total coefficients. This gives a common variance of 3,034.68.

- b. Calculate C_p , reduced AIC, and reduced BIC for Models A and B. State which model each criterion selects. Do not compare numerical values across different criteria.

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```
residual_df.Full  sigma2_hat.Full
      359.000      3034.684
```

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```
  model p    rss      cp    aic    bic
1    A 7 1098909 8.116253 2974.640 3005.948
2    B 8 1089717 7.087434 2973.532 3008.754
```

► Show the starter code

```
aic_bic_fit_gain.A    cp_fit_gain.A    aic_penalty    bic_penalty
      3.107775          3.028819        2.000000        5.913503
cp_penalty
      2.000000
```

Since smaller values are preferred, c_p and aic select Model B and bic selects model A.

- c. Compare the improvement in fit from adding `s6` with the one-parameter penalty under each criterion. Use this comparison to explain any disagreement. Why does selecting a model not establish that it is the true data-generating model?

After adding `s6`, the RSS is reduced by 9,191.51 so the Model B fits the data better. This improvement is large enough to overcome C_p and AIC penalties but not the BIC penalty.

The result depends on the sample, the models considered, and criterion being used so selecting a model does not prove that it is a true model.

Question 5: Validation and final test data

Use rows 1 through 370 of `diabetes.csv` as the training data and rows 371 through 442 as the final test data. Consider eleven nested candidate models. For $m = 0, 1, \dots, 10$, the model with m predictors contains an intercept and the first m predictors in this order:

`age, sex, bmi, bp, s1, s2, s3, s4, s5, s6`.

An analyst proposes the following procedure:

1. Fit all eleven candidate models using the training data.
2. Calculate the MSE of each model on the final test data.
3. Select the model with the smallest test MSE.
4. Report that minimum as the final estimate of prediction error.

The analyst argues that the procedure is valid because the final test data were not used to estimate the regression coefficients.

- a. Identify the first step that misuses the final test data. Explain why the minimum of eleven test MSE values is generally too favorable as an estimate of the selected procedure's future prediction error.

Step 2 is the first step that misuses the final test data. By choosing the lowest of the eleven test MSE, the values will usually make the model look better than it really is because some test errors may be low just by chance.

- b. Rewrite the analysis using five-fold cross-validation within the training data. State how the predictor count is selected, what is refitted, and when the final test data may be used.

A valid procedure is:

1. Reserve rows 371 through 442 without inspecting their responses.
2. Divide rows 1 through 370 into five folds.
3. For each predictor count, fit on four folds and calculate MSE on the remaining fold. Repeat until every fold has served once as the validation fold.
4. Average the five validation MSE values and select the predictor count with the smallest average.
5. Refit the selected model using all 370 training observations.
6. Evaluate this fitted model once on rows 371 through 442.

- c. Suppose the analyst has already examined all eleven test MSE values. What can still be reported transparently, and what additional data would be needed for a new final evaluation?

Since eleven test MSE have already been examined, the results can be reported as exploratory but the minimum cannot be considered an unbiased final test result. A new independent test dataset or future observations would be needed for a valid final evaluation.

Reference

James, Witten, Hastie, Tibshirani, and Taylor, [*An Introduction to Statistical Learning*](#), Chapters 3, 5, and 6.